Accelerated Sparse-Coding-Inspired Feedback Neural Architecture Search for Hyperspectral Image Classification

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***Abstract*— Hyperspectral images (HSIs) have spectral variability, which leads to spectral dependence in adjacent and nonadjacent regions, and this dependence is essential for the classification of regions with mixed pixels. Current neural architecture search (NAS) methods have achieved significant advantages in HSI classification, but these methods cannot capture spectral dependence in nonadjacent regions because only use feedforward connections. Meanwhile, the cost of the search process in NAS is proportional to the scale of the search space, which limits the expansion of the search space. To address these issues, we propose a sparse-coding-inspired feedback NAS (SCIF-NAS) method for HSI classification. First, we view HSI samples as sequences and introduce a feedback mechanism in NAS to model the spectral dependence of nonadjacent regions to mitigate the effects of spectral variation. Second, we design several feedforward operations according to the characteristics of HSI, to form the search space together with feedback operations. Meanwhile, a sparse-coding-inspired NAS accelerated strategy is introduced to alleviate the search time burden caused by the expansion of search space. Third, we integrate center loss with cross-entropy loss to construct a hybrid loss function that helps to obtain a better classification boundary. Finally, we conduct experiments on three popular HSI benchmarks, which show that SCIF-NAS outperforms the state-of-the-art methods in HSI classification.**

***Index Terms*— Feedback connection, hyperspectral image (HSI) classification, neural architecture search (NAS), sparse constraint.**

I. Introduction

HYPERSPECTRAL image contains rich spatial information and spectral information with dozens to hundreds of spectral bands, which reflects the physical characteristics of land cover and provides strong support for the hyperspectral image (HSI) classification. Therefore, HSI classification has been widely applied in urban development, geological exploration, agriculture, military [1], [2], and other fields. However, the acquisition of HSI is inevitably influenced by various factors, such as multipath scattering, geometric distortion, and weather conditions, leading to spatial variations in spectral features [3], [4] and the spectral dependence of HSI between adjacent and nonadjacent regions [5], [6]. These factors present significant challenges for HSI classification [7].

In the past decade, many HSI classification methods have modeled spectral dependencies based on deep learning [8], [9], [10]. Zhang and Li [11] rearranged the pixels within the patch and used 1-D convolution to extract abstract spectral features. Lin et al. [12] flattened the pixels in the adjacent region and extracted features through an autoencoder network. Duan et al. [13] proposed a method based on the fusion of dual spatial information that considered the correlation among adjacent pixels. Lv et al. [14] introduced novel operations to make the spatial–spectral information interact more closely. He et al. [15] used multiscale 3-D convolution to capture both spatial and spectral features simultaneously, while resulting in a heavy computational burden. Then, Roy et al. [16] built a model by hybrid 2-D convolution and 3-D convolution. Both Zhang et al. [17] and Cui et al. [18] proposed a lightweight 3-D convolution module to reduce the computational burden. Hong et al. [19] proposed SpectralFormer, which learned local spectral representations from multiple adjacent frequency bands rather than a single band. Furthermore, some methods attempted to establish the spatial dependence of spectral data through multimodal data. Yao et al. [20] implemented self-attention and cross-modal attention mechanisms, simultaneously capturing long-range inter-modal and intra-modal correlations between tokenized features. Hong et al. [21] effectively preserved the spatial topological structure of the studied scenes through parallel high-low resolution fusion, and they also [22] explored various multimodal fusion strategies. These methods achieve remarkable performance in the scene where the spectral variation is not significant. However, they model the spectral dependence of adjacent regions by the locality of the convolution, which ignores the spectral dependence of nonadjacent regions [23], [24].

To address the above problem, some researchers have paid attention to the recurrent neural network (RNN), which

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can model the spectral dependence of nonadjacent regions. Zhang et al. [25] presented a nonlocal spatial sequential method to find nonlocal similar structures to a given pixel. Zhou et al. [26] used a multiscanning strategy with RNN. Besides, some researchers have combined the recurrent mechanism with CNN. Liu et al. [27] constructed a multiscale alternately updated clique network. Shi et al. [28] designed a similar feedback block to enhance the correlation of nonadjacent regions. Yu et al. [29] proposed a spatial–spectral dense CNN framework with a feedback attention mechanism. Li et al. [30] implemented a spatial attention-driven recurrent feedback convolutional neural network. Moreover, Hong et al. [31] proposed SpectralGPT, which coupled spectral and spatial information by generating 3-D tokens, and used multiobject reconstruction to preserve spectral features. These methods have better robustness than the previous methods which only consider the spectral dependence of adjacent regions. However, these methods designed the architecture manually, which is time-consuming and labor-intensive. Furthermore, the manually designed architectures cannot be generalized to different HSI data with different bands, spectral ranges, and spatial resolution.

In recent years, neural architecture search (NAS) [32] has been proposed, which can automatically search high-performance network architectures according to the dataset, effectively avoiding repetitive work. For example, Chen et al. [33] applied the differentiable architecture search (DARTS) to the HSI classification task (AT-CNN). Zhang et al. [34] proposed 3-D-ANAS by designing a 3-D asymmetric CNN under the pixel-to-pixel classification framework, which was more flexible to deal with the spectral variation in adjacent regions. Then, Xue et al. [35] optimized the search space and introduced a Transformer structure based on 3-D-ANAS. To reduce the model parameters and promote spatial–spectral feature fusion, Cao et al. [36] proposed LMSS-NAS by designing an efficient multiscale search space. Wang et al. [37] proposed a NAS-guided network, which achieved fewer parameters and lower computation cost by exploiting multiscale Ghost grouped efficient channel attention (ECA) module. Although NAS-based methods can search for efficient HSI classification networks, they are faced with a dilemma: the search speed is greatly impacted by their search space and search strategy. Increasing the search space scale exponentially increases search time, which makes it extremely difficult to expand the search space.

In this article, a sparse-coding-inspired feedback NAS (SCIF-NAS) for HSI classification is proposed. The main contributions of this work include the following three aspects.

1) To alleviate the influence of spectral variation between nonadjacent regions in HSI and automatically design the neural architectures according to the characteristics of HSI, we propose a feedback NAS method that considers HSI samples as sequences and automatically searches feedforward and feedback connections in the architecture. Specifically, we extract features by feedforward connections and model the spectral dependence of nonadjacent regions by feedback connections.
2) To effectively extract spatial–spectral features, we design multiscale feedforward operations to form a large-scale search space together with the feedback operations. At the same time, we introduce sparsity constraints in the architecture search phase and introduce a sparse-coding-inspired NAS strategy to update the architecture parameters, which can alleviate the dramatic increase in search time caused by the large-scale search space.
3) To obtain a better classification boundary, we use center loss and cross entropy loss (CELoss) to construct a hybrid loss function to make the feature distribution of the same class samples more compact.

The rest of this article is organized as follows. Section II summarizes related works. Section III introduces the details of SCIF-NAS. Section IV indicates the dataset used in our experiments and the experiment environment, then analyzes the results of the experiments. Finally, our work is summarized in Section V.

II. Related Work of Neural Architecture Search

In recent years, NAS algorithms have been successful in many general computer vision tasks, such as image classification [38], object detection [39], and semantic segmentation [40]. Until now, NAS has gone through three main stages: architecture search based on evolutionary algorithm (EA), architecture search based on reinforcement learning (RL), and architecture search based on gradient. RL-based approaches [38], [39], [40], [41] usually include an RNN as a metacontroller to generate the underlying architecture. EA-based approaches [42], [43], [44] evolve a series of randomly constructed models into a better architecture. However, most RL-based and EA-based methods are computationally intensive and inefficient in the search phase. Recently, to alleviate this problem, Liu et al. [32] proposed a gradient-based NAS method called DARTS. Different from EA-based and RL-based methods that train numerous student networks, DARTS is composed of multiple cells and trains a hypernetwork in the search stage. The optimal subnetwork will be selected from the hypernetwork, which significantly reduces the search burden.

DARTS [32] searches architecture by cells, which is described as a directed acyclic graph composed of n-ordered nodes $\{N_1, N_2, \ldots, N_n\}$. There are K candidate operations $\mathcal{O} = \{o_1, o_2, \ldots, o_K\}$ on each edge (e.g., identity, max-pooling and 3×3 convolution). The intermediate node N_j and the output of the cell can be described as

$$N_j = \sum_{i=0}^{j-1} \sum_{k=1}^{K} \frac{\exp\left(a_k^{(i,j)}\right)}{\sum_{k'=1}^{K} \exp\left(a_{k'}^{(i,j)}\right)} o(N_i), \quad 2 \le j \le n-1 \tag{1}$$

$$C = \text{concat}(N_0, N_1, \ldots, N_{n-1}) \tag{2}$$

where $(\exp(a_k^{(i,j)}))/(\sum_{k'=1}^{K} \exp(a_{k'}^{(i,j)}))$ is a continuous relaxation, and $a_k^{(i,j)}$ denotes the connection weight of o_k from N_i to N_j. In this way, the task of architecture search then reduces to learning a set of continuous variables $\mathcal{A} = (\boldsymbol{\alpha}^{(0,2)}, \ldots, \boldsymbol{\alpha}^{(n-2,n-1)})$, $\boldsymbol{\alpha}^{(i,j)} = (a_1^{(i,j)}, \ldots, a_K^{(i,j)})$.

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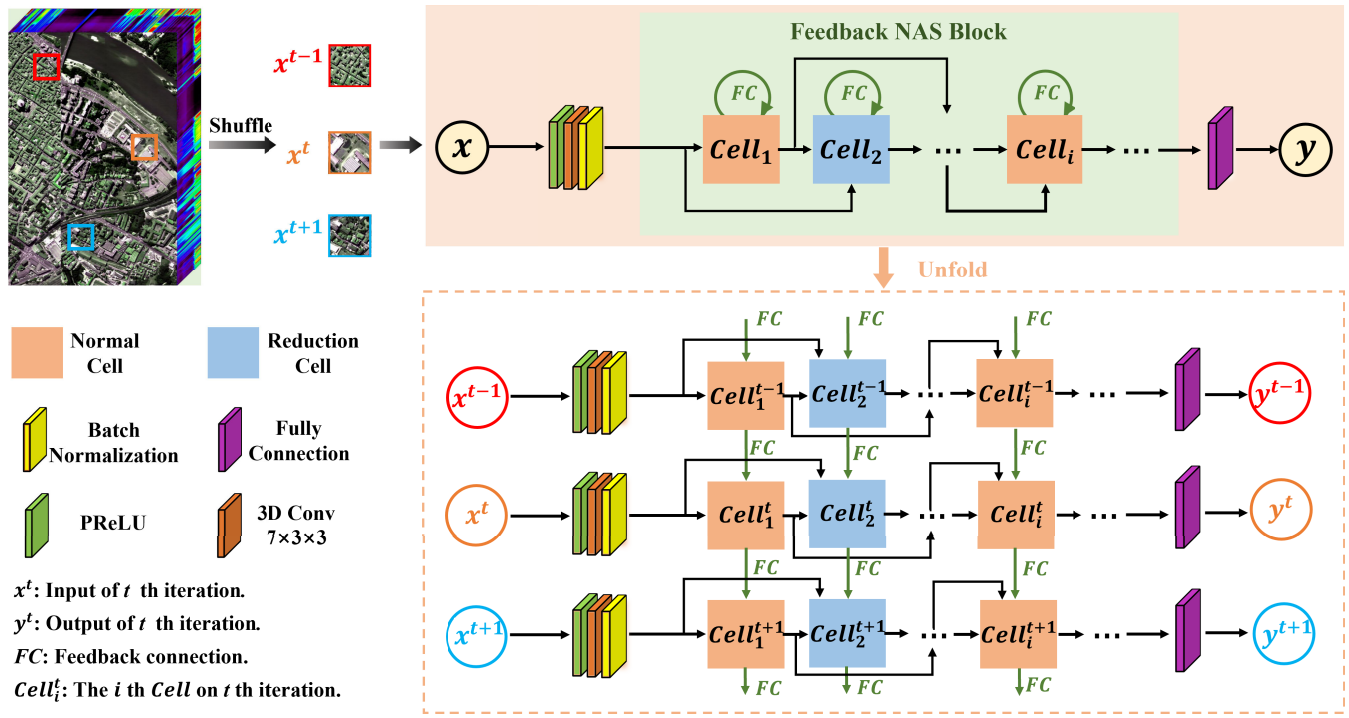


Fig. 1. Framework of SCIF-NAS. In the NAS block, $\text{Cell}_i (i = 1, 3, \ldots)$ are normal cells that set stride to one for all operations. $\text{Cell}_i (i = 2, 4, \ldots)$ are reduction cells that set stride to two for the operations adjacent to the input nodes.

The goal addressed by NAS can be described as the following two-layer optimization problem:

$$W^* = \arg\min_{W} \mathcal{L}_{\text{train}}(\mathcal{N}(W, \mathcal{A})) \tag{3}$$

$$\mathcal{A}^* = \arg\min_{\mathcal{A}} \mathcal{L}_{\text{val}}\left(\mathcal{N}\left(W^*, \mathcal{A}\right)\right) \tag{4}$$

where $\mathcal{N}$ denotes the hyper network, W represents the weight of $\mathcal{N}$, the $\mathcal{L}_{\text{val}}$ and $\mathcal{L}_{\text{train}}$ are the loss of $\mathcal{N}$ corresponding to training and validation. In the search stage, $\mathcal{A}$ and W are optimized alternately by solving (3) and (4) respectively. At the end of the search, the operation from N_i to N_j is determined by $o^{(i,j)} = \text{argmax}\, \boldsymbol{\alpha}^{(i,j)}$, which can be considered as a sparsity constraint, for any $0 \le i < n - 1$

$$||\boldsymbol{\alpha}^{(i,j)}||_0 = s = 1, \quad 0 \le i < j, \ 2 \le j \le n - 1 \tag{5}$$

where s is the sparse coefficient of $\boldsymbol{\alpha}^{(i,j)}$.

The current research generally ignores the sparsity constraint of (5) in the search stage, but obtains sparse architecture variables by post-processing. In this way, there is a gap between the hypernetwork and the optimal subnetwork, which may lead to an ineffective search process. Therefore, it was believed that every step of optimization should satisfy the constraint conditions [43], [44]. In addition, the dense hypernetwork $\mathcal{N}(W, \mathcal{A})$ generally has to use smaller depth, width, and batch sizes due to the limited memory as it covers all candidate connections in the search stage, and the search cost will increase dramatically when expanding the search space.

To liberate the limitation of search space and maintain the consistency of the networks between the search and evaluation stage, ISTA-NAS [45] was introduced to search the optimal architecture, which viewed the NAS as a sparse coding problem. Then a differentiable search is performed on a compressed low-dimensional space with the verification loss the same as the original sparse solution space, and the architecture is recovered by solving a sparse coding problem.

III. Proposed Method

In this article, a SCIF-NAS is proposed for HSI classification. First, we propose a feedback mechanism to model the spectral dependence between nonadjacent regions, as shown in Fig. 1. Second, we design a series of feedforward and feedback connections to constitute a large-scale search space for NAS. Third, to alleviate the searching time burden brought by the large-scale search space, a sparse-coding-inspired NAS strategy is introduced to accelerate the optimization process. Finally, we construct a hybrid loss to obtain a more clear classification boundary. The specific details are shown in Sections III-A–III-D.

A. Search Space

As NAS-based methods need to search over a specified search space, we design a large-scale search space, which contains two parts: feedforward operations and feedback operations, as shown in Table I, where o_1–o_{10} are feedforward operations and o_1^b–o_3^b are feedback operations. Specifically, o_1 represents the skip connection, which can effectively prevent gradient vanishing. o_2 and o_3 are average pooling and max pooling operations, respectively, used to filter out redundant information and increase the nonlinearity of the model. o_4–o_6 and o_7–o_9 are 2-D operations (Fig. 2(b), $k = 3, 5, 7$) and 3-D operations (Fig. 2(a), $k = 3, 5, 7$), respectively. As shown in Fig. 2, the beginning of the 2-D operation blocks and 3-D operation blocks is a channel attention module, which utilizes

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TABLE I
DETAILS OF THE SEARCH SPACE. o_1–o_{10} ARE FEEDFORWARD OPERATIONS. o_1^b–o_3^b ARE FEEDBACK OPERATIONS

	Name	Operations
o_1	*skip_connect*	$f(x) = x$
o_2	*avg_pooling*	$AvgPooling\,(3 \times 3)$
o_3	*max_pooling*	$MaxPooling\,(3 \times 3)$
o_4	2D_conv_3_3	$CA - [PReLU - Conv2d(3 \times 1) - Conv2d(1 \times 3) - BN] \times 2$
o_5	2D_conv_5_5	$CA - [PReLU - Conv2d(5 \times 1) - Conv2d(1 \times 5) - BN] \times 2$
o_6	2D_conv_7_7	$CA - [PReLU - Conv2d(7 \times 1) - Conv2d(1 \times 7) - BN] \times 2$
o_7	3D_conv_3_3_3	$CA - [PReLU - Conv3d(3 \times 1 \times 1) - Conv3d(1 \times 3 \times 3) - BN] \times 2$
o_8	3D_conv_5_3_3	$CA - [PReLU - Conv3d(5 \times 1 \times 1) - Conv3d(1 \times 3 \times 3) - BN] \times 2$
o_9	3D_conv_7_3_3	$CA - [PReLU - Conv3d(7 \times 1 \times 1) - Conv3d(1 \times 3 \times 3) - BN] \times 2$
o_{10}	*LKA*	$Conv2d(5 \times 5) - DilConv2d(7 \times 7, 3) - Conv2d(1 \times 1)$
o_1^b	*Avg_BACK*	$Sigmoid(GlobalAvgPool(x))$
o_2^b	*Max_BACK*	$Sigmoid(GlobalMaxPool(x))$
o_3^b	*Hybrid_BACK*	$Avg_BACK + Max_BACK$

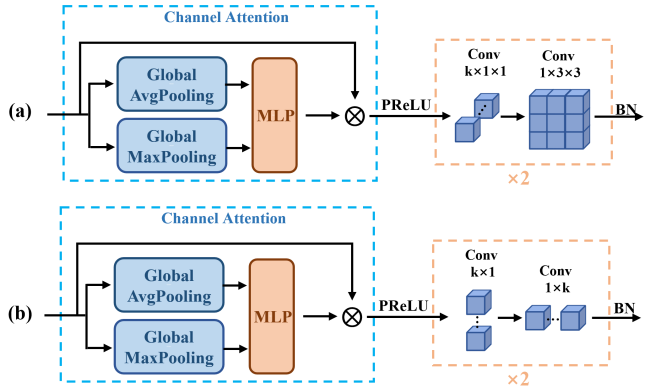


Fig. 2. Illustration of 3-D and 2-D operation sets. (a) 3-D_conv_k_3_3. (b) 2-D_conv_k_k.

global max pooling and average pooling to obtain attention weights for each channel. The end of the 2-D operation blocks and 3-D operation blocks is a depthwise separable convolution module, which reduces computational complexity while maintaining convolutional effectiveness. In particular, the 3-D depthwise separable convolution separates the $k \times 3 \times 3$ standard 3-D convolution into a $k \times 1 \times 1$ spectral convolution and a $1 \times 3 \times 3$ spatial convolution, which allows for the simultaneous extraction of spectral and spatial information while preserving spatial integrity. The LKA operation uses 5×5 2-D convolution, 7×7 dilated convolution with a dilation rate of 3, and point-wise convolution to approximate 21×21 convolution. This approximation is used to simulate a larger receptive field and capture more spatial information. Finally, the feedback operations include average feedback (global average vector), maximum feedback (global maximum vector), and hybrid feedback. Feedback connections allow the transfer of spectral feature information from the current batch to the next batch, enabling the modeling of spectral dependencies between nonadjacent regions. The detail of the feedback mechanism is elaborated in Section III-B.

B. Feedback Mechanism

Due to the influence of environmental factors during the HSI imaging process, the spectral feature is spatially distorted, which leads to the spectral dependence of HSI between adjacent and nonadjacent regions. The existing NAS-based HSI classification methods only use feedforward connection and focus on learning the spectral dependence between adjacent regions of HSI, while ignoring the spectral dependence between nonadjacent regions, which is easy to cause fuzzy classification boundaries. Inspired by the visual mechanism of the human brain, we regard HSI samples as sequences and propose a cell-wise feedback mechanism in NAS to model the spectral dependence of nonadjacent regions. To the authors' knowledge, this is the first time that the feedback mechanism is utilized in the NAS-based classification for HSI.

The feedback mechanism is directly implemented in the nodes inside the cell to pass the spectral feature vector of the current batch to the next training batch, as shown in Fig. 3, where Cell_i^t represents the ith cell in the tth training batch, and N_j^t represents the jth node in the cell during the tth training batch. During the NAS, each cell contains n nodes $\{N_0, N_1, \ldots, N_{n-1}\}$, where N_0 and N_1 are the outputs of previous cells, the other nodes are the internal node of the cell. Besides, the feedforward connections contains K candidate operations $\mathcal{O} = \{o_1, o_2, \ldots, o_K\}$, and the feedback connection contains M candidate operations $\mathcal{O}^b = \{o_1^b, o_2^b, \ldots, o_M^b\}$. When the feedback connection is not considered, the node N_j^t can be expressed as

$$N_j^t = z_j \cdot o_j, \quad 2 \le j \le n-1 \tag{6}$$

$$o_j = \left(o_1\left(N_0^t\right), \ldots, o_k\left(N_i^t\right), \ldots, o_K\left(N_{j-1}^t\right)\right) \tag{7}$$

where $o_j \in \mathbb{R}^{(j-1)K}$, and the $o_k(N_i^t)$ is the output of N_i^t by o_k, the z_j is defined by rearranging (5)

$$\begin{cases} \boldsymbol{\alpha}^{(i,j)} = \left(a_1^{(i,j)}, \ldots, a_k^{(i,j)}, \ldots, \alpha_K^{(i,j)}\right) \\ z_j = \left(\boldsymbol{\alpha}^{(0,j)}, \ldots, \boldsymbol{\alpha}^{(i,j)}, \ldots, \boldsymbol{\alpha}^{(j-1,j)}\right) \end{cases} \tag{8}$$

where $a_k^{(i,j)}$ is defined in (1), $z_j \in \mathbb{R}^{jK}$. Then, when the feedback connection is considered, the node N_j^t can be expressed as

$$\text{FC_}N_j^{t-1} = z_j^b \cdot o^b \tag{9}$$

$$N_j^t = \text{FC_}N_j^t \times (z_j \cdot o_j) \tag{10}$$

where $\text{FC_}N_j^{t-1}$ is the spectral feature vector extracted from N_j^{t-1}, the $\times$ represents channel-wise multiplication, o^b and z_j^b are defined as follows:

$$o^b = \left(o_1^b\left(N_0^{t-1}\right), \ldots, o_m^b\left(N_i^{t-1}\right), \ldots, o_M^b\left(N_{j-1}^{t-1}\right)\right) \tag{11}$$

$$\begin{cases} \beta^{(i,j)} = \left(b_1^{(i,j)}, \ldots, b_m^{(i,j)}, \ldots, b_M^{(i,j)}\right) \\ z_j^b = \left(\beta^{(2,j)}, \ldots, \beta^{(i,j)}, \ldots, \beta^{(n-1,j)}\right) \end{cases} \tag{12}$$

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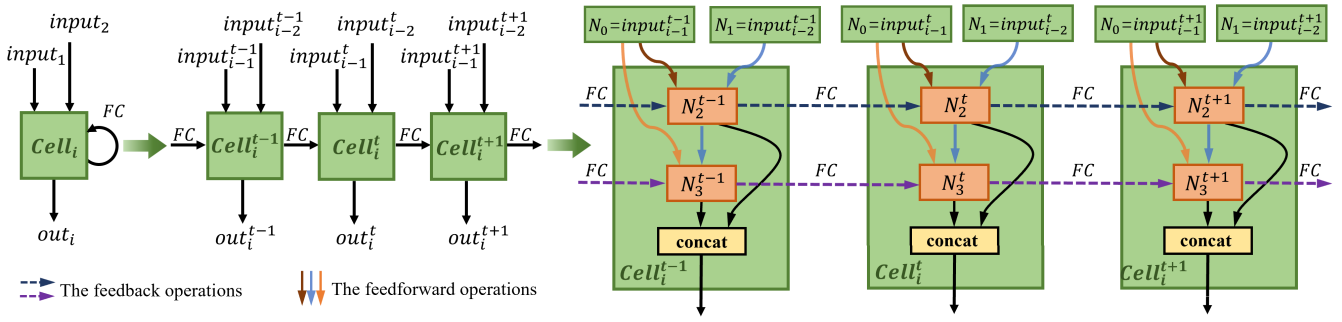


Fig. 3. Detail of feedback mechanism (for clarity of presentation, assume that each cell contains two internal nodes, and the search space contains three feedforward operations and two feedback operations). N_0 and N_1 are set equal to the outputs of Cell_{i-1} and Cell_{i-2}, N_2 and N_3 are the internal node of the cell.

$o_m^b(N_i^{t-1})$ is the output of N_i^{t-1} by operation o_m^b, the variable $b_m^{(i,j)} \in \{0, 1\}$ denotes the connection weight of o_m^b from N_i^{t-1} to N_j^t.

C. NAS Strategy

In this article, a sparse-coding-inspired NAS strategy is introduced to search for the optimal architecture, which expresses the NAS as a sparse-coding problem. A differentiable search is performed in a compressed low-dimensional space with the same validation loss as the original sparse solution space, and the architecture is recovered by solving the sparse-coding problem.

To keep the sparsity of z_j (8) and z_j^b (12) during the architecture search, a sparsity constraint is imposed on z_j and z_j^b

$$||z_j||_0 \le s, \quad \left\|z_j^b\right\|_0 \le s^b, \ 2 \le j \le n-1 \tag{13}$$

where s and s^b are sparsity coefficient, then the sparse vector z_j and z_j^b can be compressed into vector $b_j \in \mathbb{R}^{m_j}$ and $b_j^b \in \mathbb{R}^{m_j^b}$ $(m_j < jK, m_j^b < (n-2)M)$ expressed as

$$\begin{cases} b_j = \phi_j z_j + \epsilon \\ b_j^b = \phi_j^b z_j^b + \epsilon \end{cases} \tag{14}$$

where $\epsilon \in \mathbb{R}^{m_j}$ is additive noise, $\phi_j \in \mathbb{R}^{m_j \times (j-1)K}$ and $\phi_j^b \in \mathbb{R}^{m_j^b \times (n-2)M}$ denote the measurement matrix. Thus, the NAS problem converts to recover the sparse vector z_j and z_j^b from their compressed vector b_j and b_j^b, which can be regarded as solving a LASSO formulation [39]

$$\begin{cases} \min\limits_{z_j} \dfrac{1}{2}||\phi_j z_j - b_j||_2^2 + \lambda||z_j||_1 \\ \min\limits_{z_j^b} \dfrac{1}{2}\left\|\phi_j^b z_j^b - b_j^b\right\|_2^2 + \gamma||z_j^b||_1 \end{cases} \tag{15}$$

where λ and γ are the regularization coefficient. Accordingly, the NAS problem is formulated as

$$\begin{cases} z_j = \min\limits_{z} \dfrac{1}{2}||\phi_j z - b_j||_2^2 + \lambda||z||_1 \\ z_j^b = \min\limits_{z^b} \dfrac{1}{2}\left\|\phi_j^b z^b - b_j^b\right\|_2^2 + \gamma||z^b||_1 \end{cases} \tag{16}$$

Algorithm 1 Architecture Search in SCIF-NAS

Initialize: the network learning rate ξ, architecture learning rate ε, network weight W, architecture variables set B_{norm} and B_{back}.

while B_{norm}, B_{back} *not converge, for* $2 \le j \le n-1$ **do**

- Recover z_j from b_j by solving Eq.(16);
- Keep the top-s value in z_j and set others as zeros;
- Recover z_j^b from b_j^b by solving Eq.(16);
- Keep the top-s^b value in z_j^b and set others as zeros;
- Derive a sub-net $\mathcal{N}(W_{sub}, B_{norm}, B_{back})$ by z_j and z_j^b;
- $W_{sub} \leftarrow W_{sub} - \xi \nabla_{W_{sub}} \mathcal{L}_{train}(\mathcal{N}(W_{sub}, B_{norm}, B_{back}))$;
- $b_j \leftarrow b_j - \varepsilon \nabla_{b_j} \mathcal{L}_{val}(\mathcal{N}(W_{sub}, B_{norm}, B_{back}))$;
- $b_j^b \leftarrow b_j^b - \varepsilon \nabla_{b_j^b} \mathcal{L}_{val}(\mathcal{N}(W_{sub}, B_{norm}, B_{back}))$;

end

Return A sparse architecture according to z_j and z_j^b.

$$\begin{cases} B_{\text{norm}}^* = \min\limits_{B_{\text{norm}}} \mathcal{L}_{\text{val}}\left(\mathcal{N}\left(W^*, B_{\text{norm}}, B_{\text{back}}\right)\right) \\ B_{\text{back}}^* = \min\limits_{B_{\text{back}}} \mathcal{L}_{\text{val}}\left(\mathcal{N}\left(W^*, B_{\text{norm}}, B_{\text{back}}\right)\right) \end{cases} \tag{17}$$

$$W^* = \min_{W} \mathcal{L}_{\text{train}}\left(\mathcal{N}\left(W, B_{\text{norm}}^*, B_{\text{back}}^*\right)\right) \tag{18}$$

where $B_{\text{norm}} = \{b_j | 2 \le j \le n-1\}$ and $B_{\text{back}} = \{b_j^b | 2 \le j \le n-1\}$ represent the trainable parameters in the network $\mathcal{N}(W, B_{\text{norm}}, B_{\text{back}})$, z_j and z_j^b is recovered by solving (16) using iterative shrinkage threshold algorithm [46]. The detail of SCIF-NAS is shown in Algorithm 1.

D. Loss Function

CELoss is widely used in HSI classification, it focuses on learning the information between classes and it relies on the accuracy of the correct label prediction probability while ignoring the differences of other incorrect labels, which results in the learned features being scattered. Therefore, the CELoss makes it difficult to cope with the classification boundary blurring caused by the spectral variation in the HSI classification task. Thus, we introduce Center Loss [47] to assist the network parameter optimization. The Center Loss can assist in adjusting the distance between the sample and the corresponding class center, making the intra-class features

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more compact to better cope with the influence of spectral variation. The equation for calculating CELoss is as follows:

$$\text{CELoss}(x, y) = -\sum_{i=1}^{K} y_i \log(q_i) \tag{19}$$

$$y_i = \begin{cases} 1, & \text{if } i = y \\ 0, & \text{if } i \neq y \end{cases}, \quad q_i = \frac{\exp(x_i)}{\sum_{j=1}^{K} \exp(x_j)} \tag{20}$$

where K is the number of categories of the hyperspectral dataset, the actual label $y \in [1, K]$, and y_i represents the probability of each category of label y by the model according to input information. The Center Loss can effectively describe the change in the representative class, and the formula is as follows:

$$\text{CenterLoss} = \frac{1}{2} \sum_{i=1}^{m} ||x_i - c_{y_i}||_2^2 \tag{21}$$

where c_{y_i} represents the category center for all features of category y_i. Specifically, in the training process, the feature center of each category is counted out on the minibatch scale, so as to minimize the distance between all features and the feature center of the category.

The final loss is expressed as

$$\text{Loss} = \alpha \text{CELoss} + \beta \text{CenterLoss} \tag{22}$$

where α and β are used to adjust the proportion of CELoss and CenterLoss.

IV. Experimental Results

In this section, we verify the validity of the proposed method on three real HSI datasets. First, We introduce the datasets used in our experiments and the implementation environment. Second, we evaluate the classification performance to compare the proposed method with state-of-the-art methods. Third, we analyze the impact of hyperparameters on model performance. Subsequently, to verify the search efficiency of SCIF-NAS, we discuss the computational complexity and compare the search speed of sparse-coding-inspired NAS strategy with classical differentiable NAS strategy. Then, we display the details of the ablation experiment. Finally, we conduct sensitivity experiments on the loss function.

A. Datasets

Three representative HSI datasets are used: Houston University (HU), Pavia University (PU), and Pavia Center (PC). HU was provided by the 2013 IEEE GRSS Data Fusion Competition, which has 144 spectral bands and contains 15 landcover categories. Both PU and PC were acquired during a flight campaign over Pavia, Northern Italy. The number of spectral bands is 103 for PU and 102 for PC. Both datasets label nine classes. The false color composites and ground-truth maps of three datasets are shown in Fig. 4 and the detail attribute of three datasets are listed in Table II.

B. Experiment Setting

For the sample selection scheme, there are two different sampling settings in the experiments. One randomly selects 20 and ten samples from each category to construct the training set and validation set respectively, and the rest are used as the test set. The other increases the training samples to 30 samples/class, while the rest are consistent with the former. The proportion of training samples to original data is shown in Table III. Besides, to obtain stable experimental results, all results are averaged over five repeated experiments.

In the search phase, we stack three cells to search for the optimal cell structure. Meanwhile, the hyperparameters are set as follows: stochastic gradient descent (SGD) is selected as the optimizer, the initial network learning rate and architecture learning rate are set as 0.1 and 6e−3, respectively, and the cosine annealing algorithm is used to adjust the learning rate, the batch size is set to 32, and the network is iterated for 50 epochs. In the evaluation phase, we stack five cells to evaluate the performance of the optimal cell structure. In addition, the initial network learning rate is 0.025, and 500 epochs are trained. Then, the batch size, optimizer, and learning rate adjustment strategy are the same as in the search phase. All experiments are implemented on a single GTX 1080 Ti GPU, which has 10-GB memory.

C. Classification Results

We compare with six state-of-the-art HSI classification methods where the results for all comparison methods are derived from official code, containing manually designed methods (e.g., EESINet [18], LDWNet [15]), and NAS-based methods (e.g., 1-D-AT-CNN [33], 3-D-AT-CNN [33], 3-D-ANAS [34], and LMSS-NAS [36]). Three evaluation metrics including overall accuracy (OA) [48], average accuracy (AA) [48], and Kappa coefficient (K) [48] are used to measure the classification performance. The optimal structures of the normal cell and reduction cell are shown in Fig. 5. Specifically, on HU, the normal cell prefers the pooling operation. On the contrary, the normal cell is more inclined to choose the convolution operation on PU. While on PC, the normal cell has a more balanced choice between convolution and pooling. Nevertheless, the reduction cell tends to choose convolution operations on all three datasets. The difference in architecture between different datasets reflects the adaptability of SCIF-NAS to different data characteristics.

The quantitative results on HU, PU, and PC are listed in Tables IV–VI, and the visualized results are shown in Figs. 6–8. It can be seen the proposed SCIF-NAS outperforms the comparison methods on each evaluation metric and achieves state-of-the-art. For HU and PU datasets, the proposed method outperforms other methods, and for the PC dataset, the performance of SCIF-NAS is comparable to 3-D-ANAS and LMSS-NAS.

When the sampling setting is 20 samples/class. As shown in Table IV, the proposed SCIF-NAS achieves 95.90% in terms of OA on HU, while the three manually designed methods EESINet and LDWNet achieve 85.37%, and 91.31%, respectively. Compared with the four NAS-based methods 1-D-AT-CNN, 3-D-ANAS, 3-D-AT-CNN, and LMSS-NAS, SCIF-NAS improves by 37.55%, 12.75%, 12.53%, and 3.89% in terms of OA. In Table V, one can see that the proposed method performs 98.54% in OA on PU, which outperforms both manually designed and NAS-based methods. On PC as

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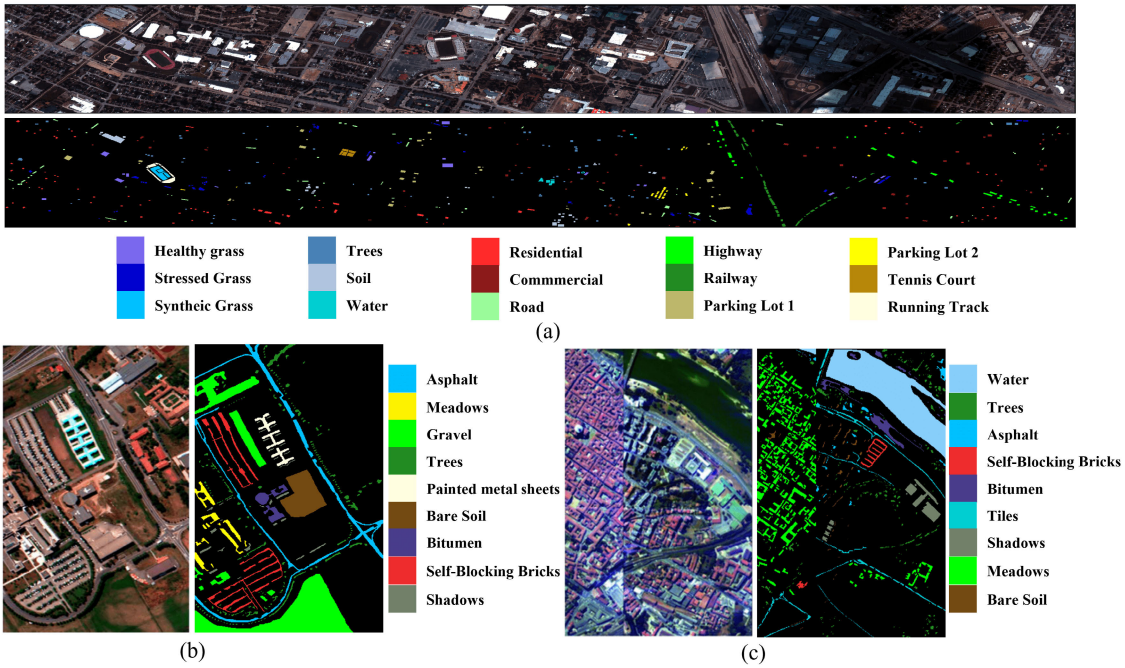


Fig. 4. False color image and ground-truth maps. (a) HU. (b) PU. (c) PC.

TABLE II
ATTRIBUTE OF THREE HYPERSPECTRAL DATASETS

Dataset	Houston University		Pavia University		Pavia Center	
Class No.	Category	No. of Samples	Category	No. of Samples	Category	No. of Samples
1	Healthy Grass	1251	Asphalt	6631	Water	65971
2	Stressed Grass	1254	Meadows	18649	Trees	7598
3	Syntheic Grass	697	Gravel	2099	Asphalt	3090
4	Trees	1244	Trees	3064	Self-Blocking Bricks	2685
5	Soil	1242	Painted metal sheets	1345	Bitumen	6584
6	Water	325	Bare Soil	5029	Tiles	9248
7	Residential	1268	Bitumen	1330	Shadows	7287
8	Commercial	1244	Self-Blocking Bricks	3682	Meadows	42826
9	Road	1252	Shadows	947	Bare Soil	2863
10	Highway	1227				
11	Railway	1235				
12	Parking Lot 1	1233				
13	Parking Lot 2	469				
14	Tennis Court	428				
15	Running Track	600				
	Total	15029	Total	42776	Total	148152
Sensor	ITRES CASI-1500		ROSIS		ROSIS	
Ground Resolution	2.5 m		1.3 m		1.3 m	
Spectral Region	360∼1050nm		0.43∼0.86μm		0.43∼0.86μm	
DataSize	349×1905×144		610×340×103		1096×715×102	
Categories	15		9		9	

listed in Table VI, it can be seen that our method achieves best in OA (99.34%) and K (91.18%), and for other methods, SCIF-NAS improves OA and Kappa by at least 0.13% and 0.18%. Meanwhile, when increasing the training samples to 30 samples/class, in terms of OA, the proposed SCIF-NAS achieves comparable results compared with both the best NAS-based methods (3-D-ANAS and LMSS-NAS) and manually designed method (LDWNet). In addition, we counted the spectral curves of some ground object categories in the highlighted box, as shown in Figs. 6(a)–8(a). It can be observed that there are spectral variations in the three HSI datasets, and the spectral variation of HU is more significant. The comparison experiment well reflects the influence of this phenomenon. The misclassified regions of the comparison method are mainly

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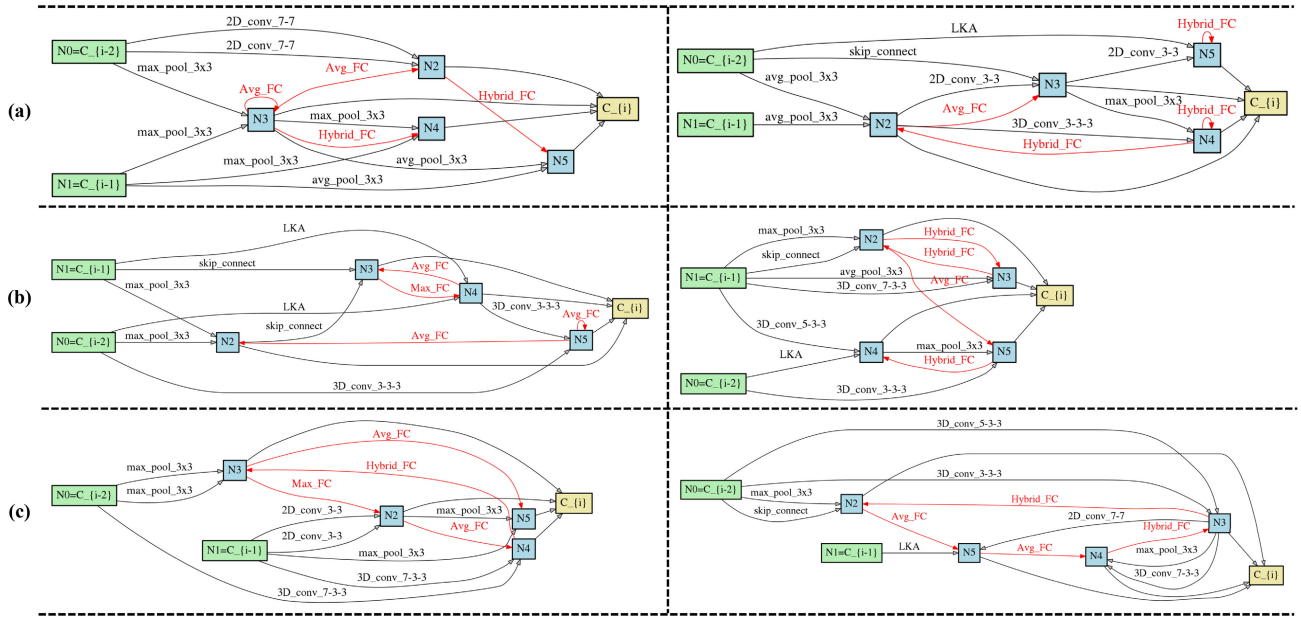


Fig. 5. Optimal structure of the normal cell (left) and reduction cell (right) on three datasets, the red arrow indicates feedback connection. (a) HU. (b) PU. (c) PC.

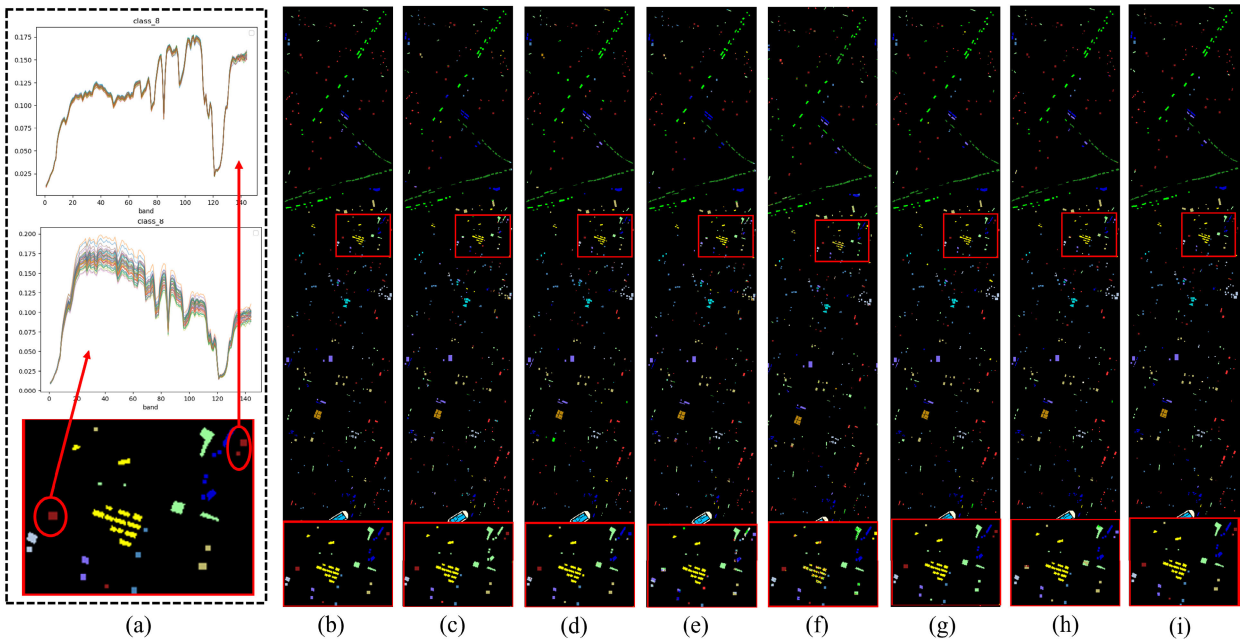


Fig. 6. Visualized results on HU dataset using 30 training samples in each class. (a) Spectral curves of the same ground object (class 8) in nonadjacent areas. (b) Ground truth. (c) EESINet. (d) LDWNet. (e) 1-D AT-CNN. (f) 3-D-ANAS. (g) 3-D AT-CNN. (h) LMSS-NAS. (i) Ours.

TABLE III
DATASET SAMPLE PARTITION

Dataset	Categories	Training Sample Partition	Validation Sample Partition	Test Sample	Training Sample Rate
HU	15	20	10	14579	1.99%
PU	9	pixel/class	pixel/class	42506	0.42%
PC	9	for training	for validation	147882	0.12%
HU	15	30	10	14429	2.99%
PU	9	pixel/class	pixel/class	42416	0.63%
PC	9	for training	for validation	147792	0.18%

concentrated in the spectral variation region, while the proposed SCIF-NAS can well alleviate the influence of spectral variation.

To summarize the experimental results on three datasets, we can find the NAS-based methods outperform manually designed methods, which proves that the NAS-based methods can find effective architectures on different datasets. It is worth mentioning that in the all compared methods, 1-D-AT-CNN only considers spectral features, making it susceptible to interference from spectral variations, such as "different object with same spectrum" and "same object with different spectrum." On the contrary, the other manually designed methods and NAS-based methods that consider spatial–spectral features are significantly better than 1-D-AT-CNN. In the proposed SCIF-NAS, we design a large-scale search space that includes feedforward operations and feedback operations.

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TABLE IV
COMPARISON EXPERIMENTAL RESULTS ON HU DATASET. LEFT: USING 20 TRAINING SAMPLES EACH CLASS. RIGHT: USING 30 TRAINING SAMPLES EACH CLASS

Sample Scheme	20 samples/class to training 10 samples/class to validation							30 samples/class to training 10 samples/class to validation						
NO.	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS
1	92.58	88.72	69.98	82.06	84.01	**97.52**	96.14	97.14	95.94	72.25	96.16	87.50	**97.69**	95.38
2	93.73	98.32	60.19	79.17	85.65	90.51	**99.47**	95.16	97.73	63.96	84.31	77.91	96.96	**99.78**
3	98.86	99.47	77.30	96.10	93.86	96.84	**100.00**	98.69	97.78	76.61	86.06	92.74	**99.10**	99.00
4	93.82	98.55	49.02	81.71	66.77	93.09	**98.72**	90.42	96.55	51.29	79.74	72.65	91.62	**99.90**
5	95.31	**95.58**	83.09	94.39	93.83	87.60	95.57	88.07	94.58	82.25	97.11	96.14	97.13	**97.79**
6	93.36	95.14	50.46	**99.32**	80.43	81.82	99.22	96.91	95.84	55.32	87.46	84.86	81.27	**99.65**
7	84.83	90.27	30.93	81.58	72.21	82.92	**94.53**	86.58	91.41	34.20	77.79	73.03	92.88	**96.70**
8	67.07	96.46	50.53	70.29	70.96	**99.27**	97.54	73.46	99.89	62.11	78.75	76.64	98.57	**99.89**
9	79.84	81.30	46.58	76.27	71.18	85.67	**95.81**	79.64	87.04	49.84	73.57	71.10	90.46	**93.91**
10	78.65	75.46	69.73	96.49	96.43	**98.34**	87.52	66.82	82.74	64.48	95.15	**96.67**	91.65	88.55
11	78.79	96.02	50.06	73.36	93.73	**98.92**	98.66	80.32	97.49	50.23	91.45	92.31	96.68	**99.62**
12	74.68	**96.35**	63.44	75.39	87.95	92.14	92.57	81.63	**98.26**	62.47	87.61	91.48	97.29	93.96
13	92.62	78.26	53.35	90.66	84.90	96.43	**97.19**	94.73	81.98	50.92	91.12	85.80	92.71	**95.94**
14	**100.00**	96.60	76.73	**100.00**	92.99	98.73	97.69	**100.00**	98.34	76.07	99.75	90.65	98.16	**100.00**
15	99.81	98.57	54.00	86.51	88.30	82.93	96.17	**99.81**	97.92	62.94	93.65	88.85	93.57	96.39
OA (%)	86.21	91.23	58.35	83.15	83.37	92.01	**95.90**	86.33	93.73	60.36	86.16	84.45	94.77	**96.65**
AA (%)	88.26	92.06	59.03	85.55	84.21	92.18	**96.45**	88.62	94.23	61.00	87.31	85.22	94.38	**97.10**
K (%)	85.09	90.90	55.18	81.81	82.07	91.36	**95.56**	85.22	93.22	57.35	85.03	83.25	94.35	**96.37**

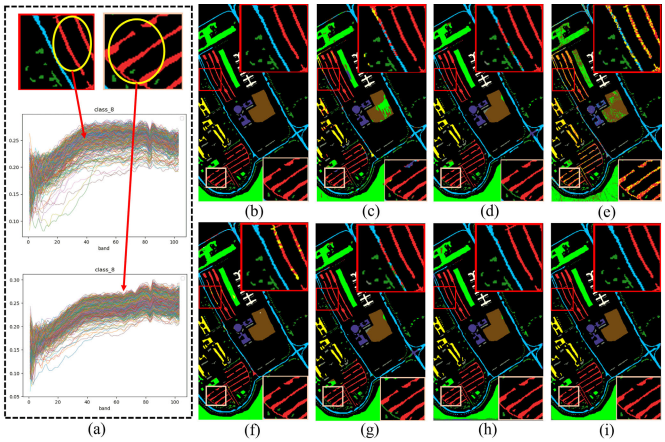


Fig. 7. Visualized results on PU dataset using 30 training samples in each class. (a) Spectral curves of the same ground object (class 8) in nonadjacent areas. (b) Ground truth. (c) EESINet. (d) LDWNet. (e) 1-D AT-CNN. (f) 3-D-ANAS. (g) 3-D AT-CNN. (h) LMSS-NAS. (i) Ours.

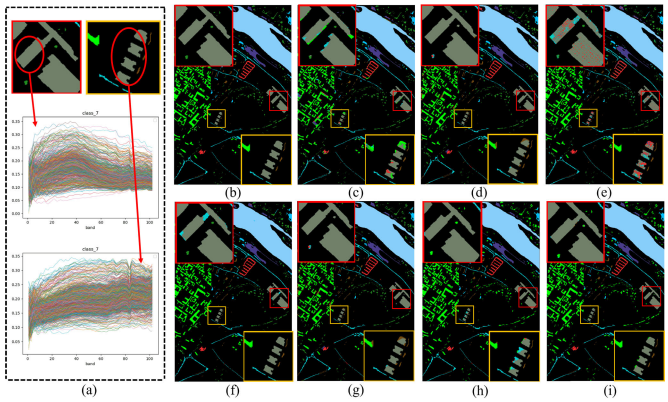


Fig. 8. Visualized results on PC dataset using 30 training samples in each class. (a) Spectral curves of the same ground object (class 7) in nonadjacent areas. (b) Ground truth. (c) EESINet. (d) LDWNet. (e) 1-D AT-CNN. (f) 3-D-ANAS. (g) 3-D AT-CNN. (h) LMSS-NAS. (i) Ours.

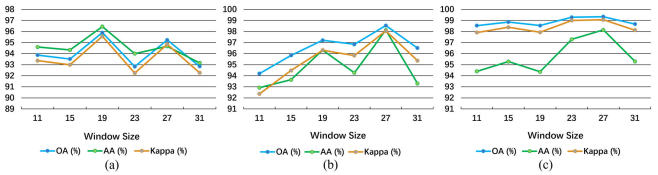


Fig. 9. Classification performance of different window sizes (11, 15, 19, 23, 27, 31) in three datasets, using 20 training samples in each class. (a) HU. (b) PU. (c) PC.

Specifically, in the feedforward operations, the multiscale spatial–spectral convolution and pooling operation provide more reliable choices for architecture search to adapt to the needs of extracting spatial–spectral features from different datasets. At the same time, the feedback operation effectively models nonadjacent region dependencies to mitigate the effects of spectral variation.

D. Hyperparameter Analysis

In general, there are many hyperparameters in HSI classification, such as window size, initial channel number, batch size, learning rate, etc., where the window size determines the spatial scale of the model input x, which directly affects the richness of spatial–spectral information. Increasing the window will bring more spatial–spectral information, however, it also increases redundancy. In this section, we explore the influence of window size on HSI classification. Fig. 9 demonstrates the performance of different window sizes in three datasets. As depicted in Fig. 9(a) for HU, the performance of SCIF-NAS fluctuates with the window size and reaches its peak at 19. For PU in Fig. 9(b), performance gradually improves as the window size increases from 11 to 27, but degrades as the window size expands further. For PC shown in Fig. 9(c), OA and Kappa only show minor fluctuations, while AA is less smooth and achieves the highest value at 27. Accordingly, we select 19 × 19, 27 × 27, and 27 × 27 spatial

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TABLE V

COMPARISON EXPERIMENTAL RESULTS ON PU DATASET. LEFT: USING 20 TRAINING SAMPLES FOR EACH CLASS. RIGHT: USING 30 TRAINING SAMPLES FOR EACH CLASS

Sample Scheme	20 samples/class to training 10 samples/class to validation							30 samples/class to training 10 samples/class to validation						
NO.	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS
1	89.69	96.44	69.69	93.21	88.24	97.48	**99.39**	93.69	99.04	76.43	93.21	89.70	99.50	**99.90**
2	79.10	97.96	76.37	99.10	90.72	99.49	**99.54**	75.93	99.59	81.90	98.50	97.92	99.50	**99.88**
3	89.44	87.63	73.43	**99.28**	92.11	94.75	98.50	85.93	94.89	74.04	**99.61**	92.31	98.47	97.68
4	97.13	97.02	90.20	**99.22**	81.27	95.91	98.80	95.38	98.38	**93.38**	98.09	71.22	97.26	97.29
5	99.65	99.55	96.54	**100.00**	93.12	98.66	99.42	99.03	99.85	96.13	**100.00**	95.12	99.63	**100.00**
6	89.38	88.71	75.48	97.36	**98.47**	92.11	95.80	92.63	95.30	81.81	**99.80**	96.96	97.99	99.34
7	76.92	94.51	88.83	**99.62**	96.14	90.64	96.43	83.67	95.69	88.23	**99.77**	95.99	95.61	99.13
8	67.35	79.17	77.59	87.35	96.84	**97.63**	96.31	76.01	90.49	74.05	96.55	94.98	97.41	**99.05**
9	99.48	95.77	96.66	98.69	79.62	95.37	**98.98**	98.61	98.08	95.65	**100.00**	80.64	97.89	98.80
OA (%)	83.74	94.00	77.59	96.53	91.16	97.17	**98.54**	83.96	97.67	81.75	97.81	93.36	98.77	**99.41**
AA (%)	87.57	92.97	82.75	96.31	90.72	95.78	**98.13**	88.99	96.73	84.62	98.39	90.54	98.14	**99.01**
K (%)	79.29	92.07	71.51	95.39	88.51	96.26	**98.06**	80.05	96.91	76.55	97.10	91.50	98.37	**99.22**

TABLE VI

COMPARISON EXPERIMENTAL RESULTS ON PC DATASET. LEFT: USING 20 TRAINING SAMPLES FOR EACH CLASS. RIGHT: USING 30 TRAINING SAMPLES FOR EACH CLASS

Sample Scheme	20 samples/class to training 10 samples/class to validation							30 samples/class to training 10 samples/class to validation						
NO.	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS	EESINet (2022)	LDWNet (2022)	1D AT-CNN (2019)	3D ANAS (2021)	3D AT-CNN (2019)	LMSS NAS (2023)	SCIF-NAS
1	99.50	99.96	99.81	**100.00**	99.56	99.95	99.98	95.21	**100.00**	99.76	99.94	99.78	99.99	**100.00**
2	91.92	97.76	80.90	94.66	87.79	96.11	**98.07**	92.93	98.58	87.90	95.49	92.48	**98.93**	97.73
3	**96.39**	81.38	89.04	96.21	82.98	94.45	94.01	**96.80**	87.83	91.16	96.73	85.81	94.80	96.02
4	97.60	92.41	73.53	**99.89**	98.85	95.68	98.64	96.33	98.23	79.72	**99.62**	97.32	98.16	91.67
5	91.76	98.10	90.05	97.60	95.83	95.33	**99.22**	79.58	**99.76**	92.10	99.69	97.02	98.28	99.55
6	88.91	92.30	95.53	**99.80**	93.22	97.60	97.39	96.21	96.46	96.47	**99.35**	95.91	97.97	97.07
7	89.49	**99.88**	84.23	97.88	94.94	98.74	99.59	90.52	99.92	85.43	97.55	95.29	99.90	**99.95**
8	96.54	**99.89**	98.17	99.25	95.12	99.47	99.57	96.15	**99.91**	97.88	99.90	95.13	99.82	99.84
9	96.27	79.04	98.89	**99.96**	85.29	94.93	96.79	95.98	96.67	98.58	**100.00**	91.84	95.42	97.22
OA (%)	96.60	98.12	96.18	99.21	96.25	98.92	**99.34**	94.57	99.29	96.79	**99.47**	96.99	99.44	99.32
AA (%)	94.26	93.41	90.02	**98.36**	92.62	96.92	98.14	93.30	97.48	92.11	**98.70**	94.51	98.14	97.67
K (%)	95.21	97.34	94.60	98.88	94.71	98.46	**99.06**	92.44	98.99	95.47	**99.24**	95.76	99.21	99.04

resolution blocks as input samples in the HU, PU, and PC datasets, respectively.

E. Model Complexity and Effectiveness of the Search Strategy

In this section, we compare the proposed method with other NAS-based methods in terms of model parameters (Params) and floating-point operations (FLOPs) and analyze the effectiveness of the search strategies. As shown in Table VII, the Params of the proposed method are higher than other NAS-based methods, while the FLOPs of 3-D ANAS are significantly higher than other methods. The Params of NAS-based methods are closely related to the search space: the compared NAS-based methods depended on the designed lightweight search spaces to accelerate the architecture search, which can not consider the large scale search space. In contrast, though we designed a larger search space with more Params to achieve a higher classification performance, we introduced the sparse-coding search strategy to accelerate the search speed.

Previous NAS-based methods are all based on DARTS search strategy and they have an inherent drawback: the scale of the search space and search speed are mutually constrained. In comparative experiments between SCIF-NAS and DARTS, we implement both search strategies with the same search space and parameter settings to find the optimal architecture on three datasets. The search times of SCIF-NAS and DARTS running for 50 epochs are shown in Table VIII. As can be seen, under the same experimental conditions, SCIF-NAS only takes 3–4 GPU min on three datasets which is less than DARTS (>65 GPU min).

TABLE VII

MODEL COMPLEXITY OF NAS-BASED METHODS ON THREE DATASETS

Model	HU		PU		PC	
	Params(M)	FLOPs(G)	Params(M)	FLOPs(G)	Params(M)	FLOPs(G)
1D AT-CNN	0.0687	0.0009	0.0725	0.0009	0.049	0.0007
3D ANAS	0.2267	3.0934	0.2156	2.0088	0.216	1.9804
3D AT-CNN	0.1091	0.029	0.0537	0.0198	0.0979	0.0261
LMSS-NAS	0.568	0.0147	0.1051	0.0192	0.0968	0.0171
SCIF-NAS	2.059	0.235	3.5745	0.603	4.0536	0.544

TABLE VIII

SEARCH TIME OF TWO SEARCH STRATEGIES ON THREE DATASETS

Dataset	SCIF-NAS	DARTS	Training Samples
HU	3.21 GPU min.	143.00 GPU min.	300 (20 each class)
PU	3.78 GPU min.	65.45 GPU min.	180 (20 each class)
PC	3.45 GPU min.	65.75 GPU min.	180 (20 each class)

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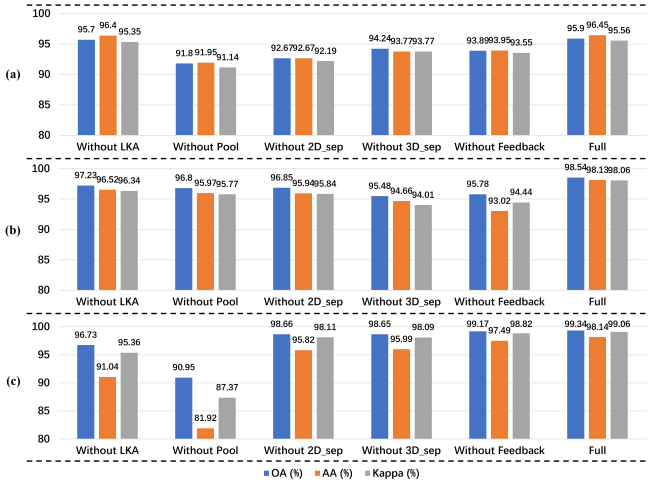


Fig. 10. Ablation experiments of different search space scales on three datasets, using 20 training samples in each class. (a) HU. (b) PU. (c) PC.

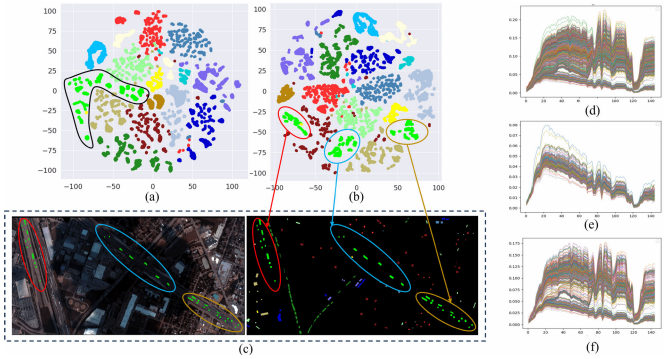


Fig. 11. 2-D t-SNE visualization of features from the HU dataset. (a) Data distributions of the labeled samples in the convolutional feature space with feedback connection. (b) Data distributions of the labeled samples in the convolutional feature space without feedback connection. (c) Region of the labeled sample (class 10, green) in the false color image (left) and the ground-truth map (right). (d) Spectra of the samples in the red circle area. (e) Spectra of the samples in the blue circle area. (f) Spectra of the samples in the brown circle area.

In addition, by comparing the search time on HU (300 training samples) with PU and PC datasets (180 training samples), we can observe that DARTS costs more time when increasing training samples, while SCIF-NAS costs stable time, which fully demonstrates that by adding sparse constraints to the architecture search process and transforming the optimization object from architecture parameters to its sparse coding, the attention on irrelevant operations can be effectively reduced, thus improving the efficiency and stability of architecture search.

F. Ablation Study

1) Effectiveness of Feedforward Operations: In this section, we implement comparative experiments of each feedforward operation on three datasets. As shown in Fig. 10, the importance of each operation is different for different datasets. On HU in Fig. 10(a), the Pool, 2-D_conv and 3-D_conv are the most important operations, when these operations are removed from the search space, the classification performance degrades significantly. However, on PU dataset, the Pool and 3-D_conv operations play a major role, as shown in Fig. 10(b). On PC dataset in Fig. 10(c), the LKA and Pool are more important than other operations. Meanwhile, we can observe the full hybrid search space has the best performance compared to others and is more balanced in terms of three evaluation metrics. The above results reflect the importance of each feedforward operation for different datasets, and verify the effectiveness of the proposed feedforward operations.

2) Effectiveness of Feedback Connections: To verify the effectiveness of the feedback connection, we implement the ablation experiments of each feedback operation on three datasets. As shown in the fifth column of Fig. 10, the addition of the feedback connection has improved the overall classification performance of the model (OA: 0.17%–2.81%, AA: 0.65%–5.11%, Kappa: 0.24%–3.62%). Notably, the feedback connection has a significant impact on AA, demonstrating that feedback connections can effectively improve the classification accuracy of the model for individual categories. Moreover, we analyze the data distributions of labeled samples in the convolutional feature space (Fig. 11). As shown in Fig. 11(a) and (b), when considering the feedback connection, the feature distribution is more concentrated than without it, and there are fewer outliers. Specifically, for example, the samples of class 10 [green dots in Fig. 11(a) and (b)] are mapped to three regions (red, blue and green circles) in convolutional feature space. Furthermore, we find the corresponding geographical location of these samples in the dataset [Fig. 11(c) and (d)] and record their spectral curves [Fig. 11(e) and (f)]. From Fig. 11(e) and (f), it can be seen that the spectral variation phenomenon is evident, where the spectra of class 10 samples in different regions are different appreciably. This indicates that it is difficult to handle spectral variation between nonadjacent regions when only using feedforward connections, which results in scattered mapping of the same category in the convolutional feature space. Conversely, when the feedback connections are considered, features of the same category in the convolutional feature space are distributed more closely and have fewer outliers [Fig. 11(a)], which fully demonstrates the ability of the proposed feedback connection to model spectral dependence between noncontiguous regions and highlight the effectiveness of feedback connections in mitigating the effects of spectral variation.

3) Effectiveness of Loss Function: The proposed method introduces the Center Loss based on CELoss, which helps to adjust the distance between the sample and the corresponding class center, making the intra-class features more compact. The influence of the Center Loss is shown in Table IX. It can be seen that the Center Loss has a positive effect on the three datasets of the ablation experiment. Specifically, the OA, AA, and Kappa on three datasets are increased by 0.67%–1.56%, 1.64%–2.24%, and 0.95%–1.68%, respectively. The Center Loss significantly improves AA, which proves that the center loss effectively improves the single-class classification accuracy.

G. Loss Function Sensitivity Analysis

In order to gain a deeper understanding of the robustness and performance of the model under different loss

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TABLE IX
ABLATION EXPERIMENTS OF THE CENTER LOSS ON THREE DATASETS, USING 20 TRAINING SAMPLES IN EACH CLASS

Dateset	Loss	OA (%)	AA (%)	Kappa (%)
HU	CE Loss	94.34	94.21	93.88
	CE Loss + Center Loss	95.90	96.45	95.56
PU	CE Loss	97.42	96.49	96.59
	CE Loss + Center Loss	98.54	98.13	98.06
PC	CE Loss	98.67	95.77	98.11
	CE Loss + Center Loss	99.34	98.14	99.06

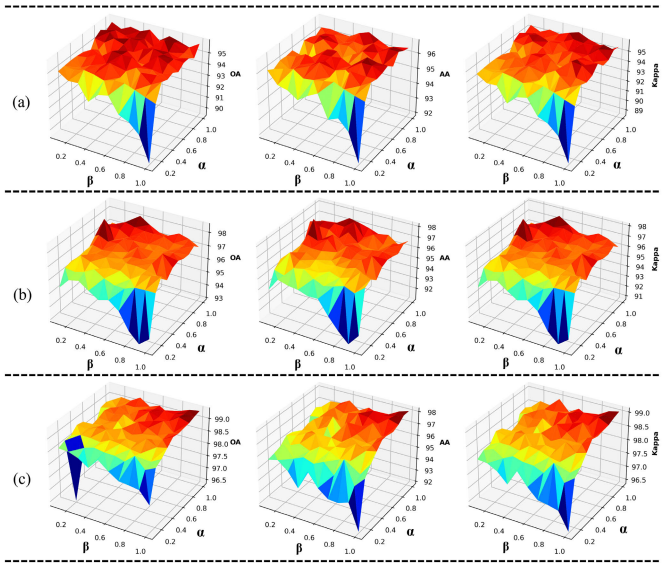


Fig. 12. Sensitivity experiments of CELoss and Center Loss on three datasets. (a) HU. (b) PU. (c) PC.

configurations, we conducted a more detailed sensitivity analysis on the parameters defined in the loss function (22). The experimental results are shown in Fig. 12. It can be observed that as the coefficient (α) of CELoss increases, the performance metrics of the model on all three datasets are improved. For Center Loss, the model performs better with lower coefficients (β) on the HU and PU datasets, and higher coefficients are better on the PC dataset. According to the experimental results, we finally determine the optimal loss coefficient on three datasets: HU ($\alpha = 0.8$, $\beta = 0.2$), PU ($\alpha = 0.75$, $\beta = 0.25$), and PC ($\alpha = 1$, $\beta = 1$).

V. CONCLUSION

In this article, we propose a SCIF-NAS method for HSI classification. Specifically, we propose a special feedback connection that the model the spectral dependence of nonadjacent regions. Then, we design several feedforward operations according to the characteristics of HSI, to form the search space together with feedback operations. Meanwhile, we use a sparse-coding-inspired NAS strategy to reduce the search cost due to the large-scale search space. The experimental results show the proposed method's advantage over other manually designed methods and NAS-based methods. Though we have achieved decent results with the proposed SCIF-NAS, the feature extraction capability of the proposed method is largely limited by the design of the operations in the search space. The current operations may not be the optimal ones, and further improvements can be made. Meanwhile, the architecture search strategy inspired by sparse coding liberates the constraint of search space scale. In future work, we will explore larger-scale general search spaces to adapt to different HSI data and task scenarios. Additionally, we aim to combine multimodal data to explore more efficient spectral-dependent modeling methods.

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